LIST OF MATHEMATICAL FORMULAE
SENARAI RUMUS MATEMATIK

Trigonometry
Trigonometri

$$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$$

$$\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$$

$$\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$$

$$\sin A + \sin B = 2 \sin\frac{A+B}{2} \cos\frac{A-B}{2}$$

$$\sin A - \sin B = 2 \cos\frac{A+B}{2} \sin\frac{A-B}{2}$$

$$\cos A + \cos B = 2 \cos\frac{A+B}{2} \cos\frac{A-B}{2}$$

$$\cos A - \cos B = -2 \sin\frac{A+B}{2} \sin\frac{A-B}{2}$$

$$\sin 2A = 2 \sin A \cos A$$

$$\begin{aligned}\cos 2A &= \cos^2 A - \sin^2 A \\ &= 2\cos^2 A - 1 \\ &= 1 - 2\sin^2 A\end{aligned}$$

$$\tan 2A = \frac{2\tan A}{1-\tan^2 A}$$

$$\sin^2 A = \frac{1-\cos 2A}{2}$$

$$\cos^2 A = \frac{1+\cos 2A}{2}$$

Differentiation
Pembezaan

$f(x)$	$f'(x)$
$\sin x$	$\cos x$
$\cos x$	$-\sin x$
$\tan x$	$\sec^2 x$
$\cot x$	$-\operatorname{cosec}^2 x$
$\sec x$	$\sec x \tan x$
$\operatorname{cosec} x$	$-\operatorname{cosec} x \cot x$

The Sum Rule
Petua Hasil Tambah

$$\frac{d}{dx}\left(u(x)+v(x)\right) = \frac{du}{dx} + \frac{dv}{dx}$$

Product Rule
Petua Hasil Darab

$$\frac{d}{dx}\left(u(x)\cdot v(x)\right) = v\frac{du}{dx} + u\frac{dv}{dx}$$

Quotient Rule
Petua Hasil Bahagi

$$\frac{d}{dx}\left(\frac{u(x)}{v(x)}\right) = \frac{v\frac{du}{dx} - u\frac{dv}{dx}}{v^2}$$

Chain Rule
Petua Rantai

$$\frac{dy}{dx} = \frac{dy}{dt} \times \frac{dt}{dx}$$

$$\frac{d^2y}{dx^2} = \frac{d}{dt}\left(\frac{dy}{dx}\right) \cdot \frac{dt}{dx}$$

Shape ***Bentuk***	**Volume** ***Isipadu***	**Surface Area** ***Luas Permukaan***
Sphere ***Sfera***	$V = \frac{4}{3}\pi r^3$	$S = 4\pi r^2$
Right circular cone ***Kon membulat tegak***	$V = \frac{1}{3}\pi r^2 h$	$S = \pi r^2 + \pi r h$
Right circular cylinder ***Silinder membulat tegak***	$V = \pi r^2 h$	$S = 2\pi r^2 + 2\pi r h$

PENILAIAN PRESTASI PELAJAR (PPP)

SECTION A [25 *marks*]

This section consists of **3** questions. Answer **all** questions.

1. (a) Let $f(x)$ be defined as

$$f(x)=\begin{cases}\dfrac{1}{x+2}, & x<2\\ k+5, & x=2\\ \dfrac{x-1}{k}, & x>2.\end{cases}$$

Determine k so that the limit of $f(x)$ as x approaches 2 exists. (4 *marks*)

(b) Evaluate $\lim\limits_{x\to-\infty}\dfrac{(x-3)(x+1)}{1-2x^2}$. (3 *marks*)

(c) Evaluate $\lim\limits_{x\to-\infty}\left(5-e^x\right)$. (2 *marks*)

2. (a) Use implicit differentiation to find $\dfrac{dy}{dx}$ for $e^y=\ln x+3y$. (5 *marks*)

(b) Given the parametric equations $x=\ln\left(t^2-1\right)$ and $y=\ln\left(t^2+1\right)^2$,

find $\dfrac{dy}{dx}$ in terms of t. Hence, evaluate $\dfrac{dy}{dx}$ at $t=-2$. (5 *marks*)

3. A curve has the equation $y=(1-x)^3+3x$. By using first derivative test,

find the relative extremums. (6 *marks*)

SECTION B [75 *marks*]

This section consists of **7** questions. Answer **all** questions.

1. Given that $z=2\left(\cos\dfrac{\pi}{3}+i\sin\dfrac{\pi}{3}\right)$ and $w=2+\sqrt{3}i$. Express $\dfrac{z}{\overline{w}}$ in the

form $a+bi$ and hence find $\left|\dfrac{z}{\overline{w}}\right|$. (5 *marks*)

2. Solve

(a) $1+\sqrt{x+3}=\sqrt{2x-1}$. (5 *marks*)

(b) $3<\left|\dfrac{x-1}{x+3}\right|$. (6 *marks*)

3. A function f is defined as $f(x)=\dfrac{1}{2}x^2-x-1$ for $x\geq 1$. Find its inverse function

and state its domain and range. (6 *marks*)

4. Given the function $g(x)=2e^{3x}-1$ and $h(x)=\frac{1}{3}\ln\left(\frac{x+1}{2}\right)$.

   (a) Show that $g^{-1}(x)$ exists. (2 *marks*)

   (b) State the domain and range of $g(x)$ and $h(x)$. (4 *marks*)

   (c) Obtain $g \circ h(x)$ and $h \circ g(x)$.

   Hence, state the relation between $g(x)$ and $h(x)$. (5 *marks*)

   (d) Sketch the graphs of $g(x)$ and $h(x)$ on the same axes. (3 *marks*)

5. Given $f(x)=\dfrac{2x^2+3}{x^2-4}$.

   (a) Find the vertical and horizontal asymptotes of $f(x)$. (8 *marks*)

   (b) Hence, sketch the graph $f(x)$. (3 *marks*)

   (c) From part (b), state the domain and range of $f(x)$. (2 *marks*)

6. (a) A curve has an equation $x^2+y^2-2y=4$.

   (i) Show that $\dfrac{dy}{dx}=\dfrac{x}{1-y}$ (3 marks)

   (ii) Determine the gradient of the curve at point $(1,3)$. (2 marks)

   (iii) Show that $\dfrac{d^2y}{dx^2}=\dfrac{5}{(1-y)^3}$ (6 marks)

   (b) Find the first derivative of $f(x)=\dfrac{2}{\sqrt{x}}$ by using the first principle. (5 marks)

7. A closed rectangular box has a base with its width $x\,cm$, height $h\,cm$ and length that is twice of the width. The total surface area of the box is $300\,cm^2$.

   (a) Show that $h=\dfrac{150-2x^2}{3x}$. (2 marks)

   (b) If the volume of the box is $V\,cm^3$. Show that $V=100x-\frac{4}{3}x^3$. (2 marks)

   (c) Find the height of the box when its volume is maximum. (6 marks)

END OF QUESTION PAPER